Giwon Baek

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Portfolio: (Link)

Personal Profile

Undergraduate researcher in robotics and computer vision with a focus on dexterous manipulation. My research focuses on generating dexterous grasp poses from 3D object point clouds with affordance awareness. I am also interested in Vision-Language-Action(VLA) models, Vision Perception and Reinforcement-Learning(RL).

Education

2020 – 2026: B.S. in Computer Science and Engineering

Sogang University, Seoul
GPA: 3.62/4.50
Relevant Coursework: Computer Vision, Basic Machine Learning, Deep Learning, Introduction to Artificial Intelligence, Operating Systems.
Key Research Interests: Dexterous manipulation, Affordance-aware manipulation, Vision-Language-Action(VLA) Models, Vision Perception, Reinforcement-Learning(RL).

Experience

Aug 2026 – Ongoing: Robotics Research Intern

Korea University RI Lab, Seoul

- Participating in RLWRD projects on vision-language-action (VLA) learning for humanoid manipulation.
- Developing data collection and training pipelines for bimanual manipulation using teleoperation demonstrations.
- Exploring VLM-based automated labeling of subtask instructions for VLA training data.

Dec 2025 – Jun 2026: Robotics Research Intern

Korea Institute of Science and Technology (KIST) HARI Lab, Seoul

- Developed affordance-aware dexterous grasp generation pipelines from 3D object point clouds using vision-language models and part-aware segmentation.
- Applied a cross-embodiment model to transfer dexterous manipulation policies to a KISTAR hand, a custom robotic hand developed at KIST.
- Collected humanoid teleoperation data for dexterous manipulation tasks using VR-based teleoperation systems.
- Constructed dexterous grasp pose datasets for the KISTAR hand using Optimization open-source models, and Isaac Gym-based validation.

Sept 2025 – Dec 2025: Like Lion AI CV Bootcamp

Like Lion, Seoul

- Completed intensive training in computer vision and deep learning.
- Improved RT-DETR-based object detection performance for autonomous driving scenarios using a Vision Mamba encoder.
- Enhanced medical image captioning performance by integrating segmentation masks with RGB breast tissue images.
- Developed a real-time deepfake detection video call application. Also received first place in the KEIT AI Life Solution Challenge.

Mar 2025 – Sept 2025: Doosan Robotics ROKEY Bootcamp

Doosan Robotics, Seoul

- Completed intensive training in robotics software, ROS 2, computer vision, and autonomous navigation.
- Developed a lane-following autonomous driving system for TurtleBot3 using a downward-facing camera, and experienced digital twin deployment from Gazebo simulation to real robots.
- Built an autonomous mobile robot pipeline using TurtleBot4, RGB-D cameras, LiDAR, YOLO-based object

detection, SLAM, and Nav2 navigation.
- Developed a voice-guided kitchen assistant robot using YOLO segmentation, PCA-based grasp orientation estimation, and MediaPipe-based hand gesture interaction.

Jun 2023 – Jul 2023 & Jan 2024 – Feb 2024: Software Engineer Intern

Learning Lab Co., Seoul

- Participated in government-funded smart manufacturing and Data voucher projects.
- Developed software pipelines for real-time manufacturing KPI collection and deployed sensor-based production monitoring systems in industrial environments.
- Collected and annotated segmentation datasets for sign language recognition and automotive defect detection tasks.
- Provided on-site deployment, maintenance, and technical support for industrial clients.

Projects

RLWRD Conveyor Demo | 2026

Korea Universisty RI Lab

- Implemented a humanoid parcel-handling task involving retrieval, flipping, and sorting.
- Collected bimanual teleoperation demonstrations and used them to fine-tune the RLDX-1 vision-language-action (VLA) model.
- Automated subtask instruction labeling for VLA training data using vision-language models.

Development of Tactile Intelligence in Robotic Hands Based on Tactile Data Learning for Manipulating Irregular Multiple Types of Objects | 2026

Korea Institute of Science and Technology (KIST) HARI Lab

- Developed an affordance-aware dexterous grasp generation pipeline from segmented 3D point clouds.
- Made a modular grasping framework with integrating Large-Language models, part-aware segmentation, and grasp pose generation.
- Generated training data using NVIDIA Isaac Gym and deployed the system on a real robot using ROS 2.

Graspium | 2026

Independent Research Seminar (Website)

- Participated in weekly paper seminars on dexterous manipulation.
- Discussed topics including dexterous grasping, multimodal VLA, reinforcement learning, and imitation learning.
- Presented papers including AffordGrasp, D(R,O) and UniMorphGrasp.

Hecto AI Challenge | 2026

Independent Team Project

- Developed a deepfake detection model based on DINOv3.
- Constructed a large-scale dataset containing 0.6 million images for model training and preprocessing.
- Awarded second place.

Korea Planning & Evaluation Institute of Industrial Technology (KEIT) AI Life Solution Challenge | 2025

Like Lion AI CV Bootcamp

- Developed a Flutter-based video call application with CNN-based real-time deepfake detection.
- Performed server-side SigLIP analysis on suspicious frames and generated explainable reports using Grad-CAM visualizations.
- Awarded first place (Chairman's Award of the KEIT Evaluation Committee).

LG Aimers 5th | 2024

LG Aimers Program

- Completed training in machine learning, deep learning and industrial applications of artificial intelligence.
- Developed an AI model to detect defective products using sensor data from the LG Display manufacturing process.
- Ranked in the top 6% among more than 750 participating teams.

Smart Manufacturing Innovation Support Program | 2024

Learning Lab Co., Ltd.

- Designed and built a counting sensor control box to measure production volume in manufacturing environments.
- Deployed sensor systems in factories and transmitted production data to office servers for KPI monitoring.
- Developed an automated pipeline to upload KPI data to government platforms through API integration.
- Contributed to the team's first-place ranking in the interim evaluation.

Data Voucher Project | 2023

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- Participated in government-funded AI dataset construction projects.
- Collected and preprocessed segmentation datasets for sign language tasks.
- Constructed defect segmentation datasets for automotive component inspection tasks.

Awards

- First Place, KEIT AI Life Solution Challenge (Chairman’s Award), 2025
- Second Place, Hecto AI Challenge, 2026

Skills

Programming: Python, C/C++
Frameworks: PyTorch, ROS 2, NVIDIA Isaac Sim, NVIDIA Isaac Gym
Operating Systems: Linux (Ubuntu)

Languages

- English: Fluent, OPIC AL
- Korean: Native
- Portuguese: Intermediate